MILITARY SPECIFICATION

POWER SUPPLY ASSEMBLY, NIXIE

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of electronic assembly known as Nixie Power Supply.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-F-13926 -	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232 -	Parts, Equipment and Tools for Army Materiel: Packaging and Packing of
MIL-1-45607 -	Inspection Equipment, Acquisition, Disposition and Maintenance of

STANDARDS

MILITARY

MIL-STD-105 -	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 -	Quality Assurance Terms and Definitions
MXL-STD-810 -	Environmental Test Methods

FSC: 1240

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS (See 6.4)

US Army Armament Research and Development Command (ARRADCOM)

11750377 - Nixie Power Supply Assembly

PACKAGING DATA SHEET

11750377 - Packaging of Nixie Power Supply Assembly 11750377

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Fabrication. - The power supply shall be manufactured in accordance with Drawing F11750377 and drawings/specifications pertaining thereto, as specified in the contract.

3.2 General Specification. - The power supply unit shall meet the following requirements, where applicable, of MIL-F-13926:

(a) Dimensions and tolerances
(b) Part identification and marking

3.3 Performance. -Unless otherwise specified, the power supply unit shall meet the performance requirements of this specification at standard ambient temperatures between +60 degrees and +90 degrees Fahrenheit (F).

3.3.1 Loads, power and signals. -The power supply unit shall perform as specified herein when the loads, auxilary equipment, power and signals are applied as specified in Table I.

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TABLE I

ITEM	CONDITION	CHARACTERISTICS	CONNECTIONS
1.	LOADS		CONNECTED BETWEEN THE FOLLOWING TERMINALS:
1.1	RESISTOR	28 Kilohms ± 1%	J1-A and J1-D
2.	POWER SOURCE	+15 ± 1.2 Vdc	J1-C (+) and J1-D (-)

3.3.1.1 Output voltage and ripple. - The output voltage between J1-A(+) and J1-D(-) shall be 214 ± 36 VDC, and the maximum ripple voltage shall be 15 volts peak-to-peak when the power source of Table I, Item 2 is applied with the indicated polarity over the range specified therein.

3.3.1.2 Input current. - The maximum currect supplied to terminal J1-C shall be 260 milli-amperes when the load and power are applied as specified.

3.3.2 Environmental.

3.3.2.1 Storage temperature. - The power supply unit shall meet the requirements of 3.3.1 at ambient temperature (+60°F to +90°F) after exposure and thermal stabilizatlon at -80°F and +160°F.

3.3.2.2 Operating temperatures. - The power supply unit shall meet the requirements of 3.3.1 while thermally stabilized at -40°F and +125°F, subsequent to which it shall meet the above requirements at ambient temperature (+60°F to +90°F).

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3.3.2.3 Vibration. -The power supply unit shall show no physical failure and shall meet the requirements of 3.3.1 after being exposed to the following vibratory conditions:

(a)	Double Amplitude	0.5 inch maximum
(b)	Acceleration	4g maximum
(c)	Crossover frequency	12.5 Hertz (Hz)
(d)	Sweep time	5 to 500 to 5 Hz in 15 minutes. Two sweeps in each of the three mutually perpendicular axes.

3.3.2.4 Shock. -The power supply unit shall meet the requirements 3.3.1 after being exposed to a total of six shock impulses, one in each direction of the three mutually perpendicular axes. Each shock impulse shall be a sawtooth with a time duration of six milliseconds. The peak amplitude shall be 100g.

3.4 First Article Inspection. -This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.5 Workmanship. -The power supply unit shall meet the workmanship requirements of MIL-F-13926 and applicable specifications as cited on Drawing 11750377.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. -Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as other wise specified in the contract or order, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. In addition, the Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions. -The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

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4.2 Classification of Inspections. -Unless otherwise specified herein or in the contract, the following types of inspection shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inpsection

4.3 First Article Inspection

4.3.1 Submission. -The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provision of 4.3.2. Unless otherwise specified, the sample for first article approval tests shall consist of three power supplies plus three each of all components and subassemblies.

4.3.2 Inspections to be performed. -First article assemblies shall be subjected to the examinations and tests specified in Tables II,III and IV herein. The Government reserves the right to subject the first article assemblies, components and subassemblies to any or all of the examinations and. tests specified herein and to any or all requirements of the applicable drawings.

4.3.3 Rejection. -If any assembly, component or subassembly fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly, component, or subassembly in the sample to comply with any of the stated requirements.

4.4 Quality Conformance Inspection

4.4.1 Inspection lot formation. -Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with “submission of product” provisions of MIL-STD-105.

4.4.2 Examination. -The examinations listed herein for Classification of Defects shall be performed on inspection lots as defined in 4.4.1 of this specification.

4.4.2.1 Components and subassemblies. -All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality

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Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.4.2.2 Final acceptance inspection. -Subsequent to first article approval, examination and test related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105, Inspection level II, and the sampling plans specified in Tables II, III, and IV herein. Examination and test for packaging and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables II, III and IV shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

4.4.2.3 Functional tests. -The requirements and tests in Table II shall be inspected on a 100% basis. All examinations and tests shall be conducted at the standard ambient temperature (+60° to +90°F).

TABLE II - Classification of Defects
(100 percent inspection)

CLASS	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURES
CRITICAL	NONE		
MAJOR:			
101.	Output voltage and ripple	3.3.1.1	4.6.2
102.	Input current	3.3.1.2	4.6.3

4.4.3 Acceptance and rejection. -Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.5 Special sampling

4.5.1 General. -One power supply shall be randomly selected as a sample FROM 100 produced or from each month's production, whichever comes first. The sample shall meet the requirements and tests in Table III.

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TABLE III

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURES
301	Fabrication	3.1	Applicable drawings Visual
302	General Specification	3.2	MIL-F-13926

4.5.2 Environmental. -Three power supply units shall be selected at random as samples from each 50 produced or from each month's production, whichever occurs first. The samples shall meet the requirements and tests in Table IV. and shall then meet the requirements and tests in Table II.

TABLE IV

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURES
303	Storage temperature	3.3.2.1	4.6.4.1
304	Operating temperature	3.3.2.2	4.6.4.1
	Vibration	3.3.2.3	4.6.4.2
	Shock	3.3.2.4	4.6.4.3

4.5.3 Failure of sample. -Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor. and the resubmitted item has been approved (see 4.4.3).

4.5.4 Inspection Equipment. -Requirements for inspection equipment to perform examinations and tests are specified herein and in specific contract clauses as applicable. See Section 6 of this specification for general provisions applicable to inspection equipment designs.

4.5.5 Government furnished inspection equipment. -Where the contract provides for government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.6 Contractor furnished inspection equipment

4.5.6.1 Test equipment. -In conjunction with 4.5.6, the standard test equipment listed in Table V shall be utilized in performing the applicable tests of 4.6.

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TABLE V

	NOMENCLATURE	DESCRIPTION
1.	DC Power Supply	Plus 15 ± 1.2 V Ripple: 50 MV pp Max
2.	DC Voltmeter	Range: 0 to 300 Vdc Accuracy: ± 1%
3.	DC Milliammeter	Range: 0 to 500 ma Accuracy: ± 5%

4.6 Methods of inspection

4.6.1 Test conditions. -Unless otherwise specified, the test conditions shall be in accordance with the "General Requirements" of MIL-STD-810.

4.6.2 Output voltage and ripple. -Apply the voltage specified in 3.3.1.1, observe and measure the output voltage and ripple voltage present between pins "A" and "D" of connector 11728953. The output shall be as specified in 3.3.1.1.

4.6.3 Input current. -Apply the voltage specified in 3.3.1.1, observe andmeasure the input current. The input current shall be as specified in 3.3.1.2.

4.6.4 Environmental

4.6.4.1 Storage and operating temperatures. -The power supply shall be temperature tested in accordance with Test Method 501.1, Procedure I and Test Method 502.1, Procedure I of MIL-STD-810 except, the power supply shall be exposed to the storage temperatures of + 160°F and -80°F for a minimum of 5 hours. In addition, the following conditions shall also apply:

a. During thermal stabilization at -40°F and +125°F operational temperatures, the power supply shall be energized and subjected to the tests of 4.6.2 and 4.6.3.

b. Subsequent to temperature testing and upon return to thermal stabilization at ambient temperature, the power supply shall be energized and again subjected to the tests of 4.6.2 and 4.6.3.

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The power supply shall conform to the requirements of 3.3.2.1 and 3.3.2.2.

4.6.4.2 Vibration. -With the power supply firmly mounted to the vibration table, perform the following:

a. Test in acordance with Method 514.2, Procedure VIII of MIL-STD-810 for category (f) equipment, (Tracked Vehicles), except the vibratory conditions of 3.3.2.3 shall apply.

b. Sweep the frequency from 5 to 500 Hz and back to 5 Hz in a period of 15 minutes. Perform two sweeps in each of the three mutually perpendicular axes.

c. Following vibration cycling examine the power supply for structural failures, loose hardware, wire breakage, and abrasion.

d. Place the power supply in operation and perform the tests specified in 4.6.2 and 4.6.3.

The power supply shall meet the requirements of 3.3.2.3.

4.6.4.3 Shock. -With the power supply unit rigidly mounted to the shock MACHIne table, perform the following:

a. Test in accordance with Method 516.2, Procedure I, of MIL-STD-810, except that the power supply unit shall be subjected to one shock in each direction of the three mutual perpendicular axes and the shock impulse shall be a sawtooth with a duration of six milliseconds and a peak amplitude of 100g.

b. After exposure to shock, perform a visual-tactile examination for damage and loose parts.

c. Place the power supply unit in operation and perform tHE test specified in 4.6.2 and 4.6.3.

The power supply unit shall meet the requirements of 3.3.2.4.

5. PACKAGING

5.1 Packaging, packaging and marking. -Packaging, packing and marking shall be in accordance with Packaging Data Sheet 11750377. The level of protection shall be as specified in the contract.

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6. NOTES

6.1 Intended use. -The Nixie Power Supply is intended for use on the Laser Range Finder AN/VVS-1, which provides 200 VDC for biasing the display Nixie tubes of the command Display Indicator Unit.

6.2 Ordering data. -Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Quantity required and delivery schedules.

c. Applicable packaging data sheet number (see 5.1).

d. Selection of applicable levels of preservation, packaging and packing.

e. Applicable stock number.

f. Requirements for First Article Submission, including data requirements for certificates of conformance for materials.

6.3 Inspection equipment design

6.3.1 Government design. -Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the Technical Data Package (TDP).

6.3.2 Contractor design. -The contractor shall design and supply inspection equipment compatible with the “Methods of Inspection” specified in 4.6 of this specification and with the component inspection procedures specified in “Examination” and “Test Facilities” requirements of MIL-F-13926. Since tolerance of test eqqipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials,dimensions shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

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6.3.3 Submission of designs for approval. -Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs and the required delivery schedule as stipulated in the contract. Submit designs as required to Commander, US Army Armament Research and Development Co-remand, ATTN: DRDAR-QAF-1, Dover, N.J. 07801.

When the contractor submits inspection equipment designs to the Government for approval he shall give the following information in his letter of transmittal:

a. The contract number.
b. The contract item (name, model number, etc.)
c. The designs remaining to be submitted and the expected date of submittal.

6.4 Drawings. -Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

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